# VERIFIED CREDENTIAL IDENTIFIER

## VCI

*A Neutral Trust Primitive for the Global Credential Economy*

**Technical Specification • ATS Integration Architecture • Governance Framework**

**Version № 1.0**

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**EXECUTIVE SUMMARY**

The Verified Credential Identifier (VCI) is a globally unique, cryptographically addressable pointer that functions as an infrastructure layer protocol of the credential economy. It is a mathematical hash of an issuer's Decentralized Identifier (DID), a credential schema, an issuance nonce, and a timestamp, VCI resolves to a specific W3C Verifiable Credential instance to state confirms to either revoked or unknown, without exposing private contents and without requiring the original issuer to be contacted. The critical property that makes VCI the correct architectural response to signal propagation in credentialing platforms is its passivity. Itt does not evaluate, adjudicate, or interpret. It transmits one bit of mathematically verified truth. The relationship between a credential signal's claims and commitments is resolved not by auditing issuers but by cryptographically verifying the transmission layer of signal architecture.

## 1 VCI: The Mathematical Definition

A VCI is a passive string defined as a deterministic hash of four inputs:

| VCI Component | Definition and Function |
|---|---|
| **Issuer DID** | The issuer's Decentralized Identifier, a URL-based identifier resolved against the Verifiable Data Registry to retrieve the issuer's current public key without contacting the issuer directly |
| **Credential Schema** | A pointer to the structured data template defining the credential's fields, ensuring the VCI resolves to the correct credential type |
| **Issuance Nonce** | A cryptographically random value generated at issuance, ensuring that two credentials of the same type issued to the same holder by the same issuer produce distinct VCIs |
| **Timestamp** | The ISO 8601 issuance timestamp, creating the enrollment-time snapshot that allows the holder to prove what was committed at the moment they received the credential |

VCI changes credential verification by embedding trust directly in the credential itself. Instead of requiring external validation services or intermediary verification steps, authenticity is confirmed directly when the credential is presented. Employers do not need to contact the issuer for confirmation. The credential is simply verified at the point of use.

This makes verification more portable and scalable across labor markets, because of this reduced reliance on centralized verification services and proprietary verification systems.

**The Dumb Pointer Principle**

VCI is a pointer, not a platform. It resolves to a state (valid/revoked/unknown) without containing, transmitting, or interpreting any private data. A VCI cannot tell an employer whether a candidate is qualified. It tells them one thing: the cryptographic proof of the credential's existence and validity checks out. The qualification judgment is the employer's. The verification is VCI's. These must never be conflated.

## 7 Relationship to the Complete Governance Stack

VCI is the technical primitive of a three-layer governance architecture. Each layer addresses a distinct problem. No layer substitutes for the others.

| Layer | Component | Function in the Stack |
|---|---|---|
| **Detection** | Runtime Divergence Analysis (RDA) | Identifies and documents the gap between institutional capability representations and observable execution architecture.. Produces primary-source, self-authenticating governance findings. |
| **Attribution** | Digital Credential Operational Trust Infrastructure (DCOTI) | Makes issuer operational significance commitments explicit, attributable, cryptographically signed, and permanently queryable. Converts marketing claims into institutional commitments against which RDA findings can be referenced. The neutral registry. |
| **Verification** | Verified Credential Identifier (VCI) | Makes credential validity cryptographically verifiable by any compliant system without institutional cooperation. Works prospectively. Closes the signal gap at the architectural level. Enables output to employer systems for signal transmission. |

The signal propagation finding hat motivated this body of work exists because none of these three layers were  operational.

1. RDA documents the gap.
2. DCOTI attributes the commitment.
3. VCI closes the gap.

Together, the three layers make the gap structurally impossible to maintain: issuers must commit specifically in DCOTI, those commitments are RDA-auditable against observable execution, and VCI ensures the execution actually transmits a verifiable signal that employers can independently confirm.

**The Enrollment-Time Snapshot is Already in the Specification**

The W3C VC Data Model v2.0's validFrom property and VDR key resolution history natively create the enrollment-time snapshot. When a holder receives a VCI-enabled credential, the issuer's commitment at that moment is cryptographically sealed in the signature and timestamp. Any subsequent change to the issuer's marketing surface does not alter the signed credential. The holder can prove what was committed at issuance without DCOTI rebuilding this feature.

### Prior Art and Publication Notice

This document establishes the original technical specification of the Verified Credential Identifier (VCI) as a neutral trust primitive for the digital credential economy. The mathematical definition (DID hash + schema + nonce + timestamp), the five neutrality boundaries, the five-layer architecture, the ATS integration flow, the VALID=1 binary output protocol, and the governance model described herein are original intellectual contributions of the author.

VCI is designed as the technical implementation layer for the Digital Credential Operational Trust Infrastructure (DCOTI) framework (Mohamed S. Mohamed, May 2026) and is compatible with the Runtime Divergence Analysis methodology (Version 1.0, May 2026) as its signal verification layer. The three documents, RDA Specification, DCOTI Framework, and this VCI Specification, together constitute a complete governance stack for credential representational integrity in the digital economy. Author: Mohamed S. Mohamed • May 2026